Experiment 7

Random Forest

Aim:

To implement the Random Forest algorithm for classification tasks and evaluate its performance using appropriate classification metrics.

Theory:

A Random Forest is a supervised ensemble learning algorithm used for both classification and regression tasks. It builds multiple decision trees during training and combines their outputs to produce a more accurate and stable prediction. Instead of relying on a single tree, Random Forest aggregates the results of many trees to reduce overfitting and improve generalization.

The main components of a Random Forest are:

1. Decision Trees: Individual models trained on different subsets of data.
2. Bootstrap Samples: Random subsets of the original dataset selected with re-placement.
3. Feature Randomness: A random subset of features is considered at each split to ensure diversity among trees.

Each tree in the forest makes a prediction, and the final output is determined by majority voting (for classification) or averaging (for regression).

A Random Forest works by constructing multiple decision trees using different bootstrap samples of the training data. For each tree, at every node, only a random subset of features is considered when selecting the best split. This randomness ensures that the trees are diverse and not highly correlated. The splitting criteria commonly used include Gini Index and Entropy. The Gini Index measures the impurity of a node, representing the probability of incorrectly classifying a randomly chosen sample if it were labeled according to the class distribution in the node. The Gini Index is calculated as:

$$\text{Gini}(S) = 1 - \sum_{i=1}^{n} p_i^2$$

Lower Gini values indicate purer nodes, with a value of zero when all samples belong to one class. However, it is slightly biased toward dominant classes.

Entropy, on the other hand, measures the amount of uncertainty or disorder in a node and is used to calculate Information Gain and is computed as:

$$\text{Entropy}(S) = -\Sigma\, p_i \log_2(p_i)$$

Entropy is zero for a perfectly pure node, with higher values indicating greater disorder. It is sensitive to small changes in class distribution and often produces balanced and informative splits. Once all trees are trained, predictions are made by combining the outputs of individual trees. In classification, the class predicted by the majority of trees is selected, while in regression, the average of all tree predictions is taken as the final output. By combining multiple trees, Random Forest reduces variance, minimizes overfitting, and provides higher accuracy compared to a single decision tree.

Merits of Random Forest

1. **High Accuracy & Robustness** – By combining multiple decision trees, Random Forest reduces overfitting and usually gives better accuracy than a single decision tree.

2. **Handles Large & Complex Data** – It works well with large datasets and can handle both numerical and categorical features effectively.

3. **Feature Importance Estimation** – Random Forest can measure the importance of each feature, helping in feature selection and understanding the model.

Demerits of Random Forest

1. **Computationally Expensive** – Training many trees requires more time and memory compared to simpler models.

2. **Less Interpretability** – The ensemble nature makes it difficult to interpret compared to a single decision tree.

3. **Not Ideal for Real-Time Prediction** – Due to multiple trees, prediction can be slower, making it less suitable for time-critical applications.

CODE FOR BINARY CLASSIFICATION

INPUT: Import Libraries

```
import pandas as pd

import numpy as np

from sklearn.model_selection

import train_test_split from sklearn.preprocessing

import LabelEncoder

from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report import

matplotlib.pyplot as plt

import seaborn as sns

print("Libraries imported successfully\n")
```

OUTPUT:

Libraries imported successfully

INPUT: Load Dataset

```
df = pd.read_csv("bank_marketing.csv", sep=';')

print("Dataset loaded successfully")
```

OUTPUT:

Dataset loaded successfully

INPUT: Check The Data Shape

```
print("Dataset shape:", df.shape)
```

OUTPUT:

Dataset shape: (41188, 21)

INPUT: Display First 5 rows

```
print("\nFirst 5 rows:") print(df.head(), "\n")
```

OUTPUT:

```
First 5 rows:
   age        job  marital    education  default housing loan    contact  \
0   56  housemaid  married     basic.4y       no      no   no  telephone
1   57   services  married  high.school  unknown      no   no  telephone
2   37   services  married  high.school       no     yes   no  telephone
3   40     admin.  married     basic.6y       no      no   no  telephone
4   56   services  married  high.school       no      no  yes  telephone

  month day_of_week  ...  campaign  pdays  previous     poutcome emp.var.rate  \
0   may         mon  ...         1    999         0  nonexistent          1.1
1   may         mon  ...         1    999         0  nonexistent          1.1
2   may         mon  ...         1    999         0  nonexistent          1.1
3   may         mon  ...         1    999         0  nonexistent          1.1
4   may         mon  ...         1    999         0  nonexistent          1.1

   cons.price.idx  cons.conf.idx  euribor3m  nr.employed   y
0          93.994          -36.4      4.857       5191.0  no
1          93.994          -36.4      4.857       5191.0  no
2          93.994          -36.4      4.857       5191.0  no
3          93.994          -36.4      4.857       5191.0  no
4          93.994          -36.4      4.857       5191.0  no

[5 rows x 21 columns]
```

INPUT: Dataset Information

print("Step 3: Dataset

information")

print(df.info(), "\n")

OUTPUT:

Data columns (total 21 columns):

#	Column	Non-Null Count	Dtype
0	age	41188 non-null	int64
1	job	41188 non-null	object

2 marital 41188 non-null object

3 education 41188 non-null object

4 default 41188 non-null object

5 housing 41188 non-null object

6 loan 41188 non-null object

7 contact 41188 non-null object

8 month 41188 non-null object

9 day_of_week 41188 non-null object

10 duration 41188 non-null int64

11 campaign 41188 non-null int64

12 pdays 41188 non-null int64

13 previous 41188 non-null int64

14 poutcome 41188 non-null object

15 emp.var.rate 41188 non-null float64

16 cons.price.idx 41188 non-null float64

17 cons.conf.idx 41188 non-null float64

18 euribor3m 41188 non-null float64

19 nr.employed 41188 non-null float64

20 y 41188 non-null object

INPUT: Categorial Encoding

```
df_encoded = df.copy() le = LabelEncoder()

for col in df_encoded.columns:

    if df_encoded[col].dtype == 'object':

        df_encoded[col] =

        le.fit_transform(df_encoded[col])

        print(f"Encoded column: {col}")

print("\nCategorical encoding completed\n")
```

OUTPUT:

```
Encoded column: job

Encoded column: marital

Encoded column: education

Encoded column: default

Encoded column: housing

Encoded column: loan

Encoded column: contact

Encoded column: month

Encoded column: day_of_week

Encoded column: poutcome

Encoded column: y

Categorical encoding completed
```

INPUT: Check the Shape of Features and Target

```
X=df_encoded.drop('y', axis=1)
```

```
y = df_encoded['y']

print("Features shape:", X.shape)

print("Target shape:", y.shape, "\n")
```

OUTPUT:

Features shape: (41188, 20)

Targetshape: (41188,)

INPUT: Split the Data

```
X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.2, random_state=42, stratify=y)

print("Training set shape:", X_train.shape)

print("Testing set shape:", X_test.shape, "\n")
```

OUTPUT:

Training set shape: (32950, 20)

Testing set shape: (8238, 20)

<u>GINI</u>

INPUT: Initialization Of Random Forest Model

```
rf_model = RandomForestClassifier( n_estimators=200, max_depth=15, min_samples_split=5,

    min_samples_leaf=2, random_state=42, n_jobs=-1 )

print("Random Forest model initialized\n")
```

OUTPUT:

Random Forest model initialized

INPUT: Predictions

```
y_pred = rf_model.predict(X_test)

print("Predictions completed\n")
```

OUTPUT:

Predictions completed

INPUT: Check The Accuracy

```
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy of Random Forest model:", accuracy, "\n")
```

OUTPUT:

```
Accuracy of Random Forest model: 0.9225540179655256
```

INPUT: Evaluation Of Model Performance

```
from sklearn.metrics importconfusion_matrix

cm = confusion_matrix(y_test, y_pred)

print("Confusion Matrix:\n")

print(cm)
```

OUTPUT:

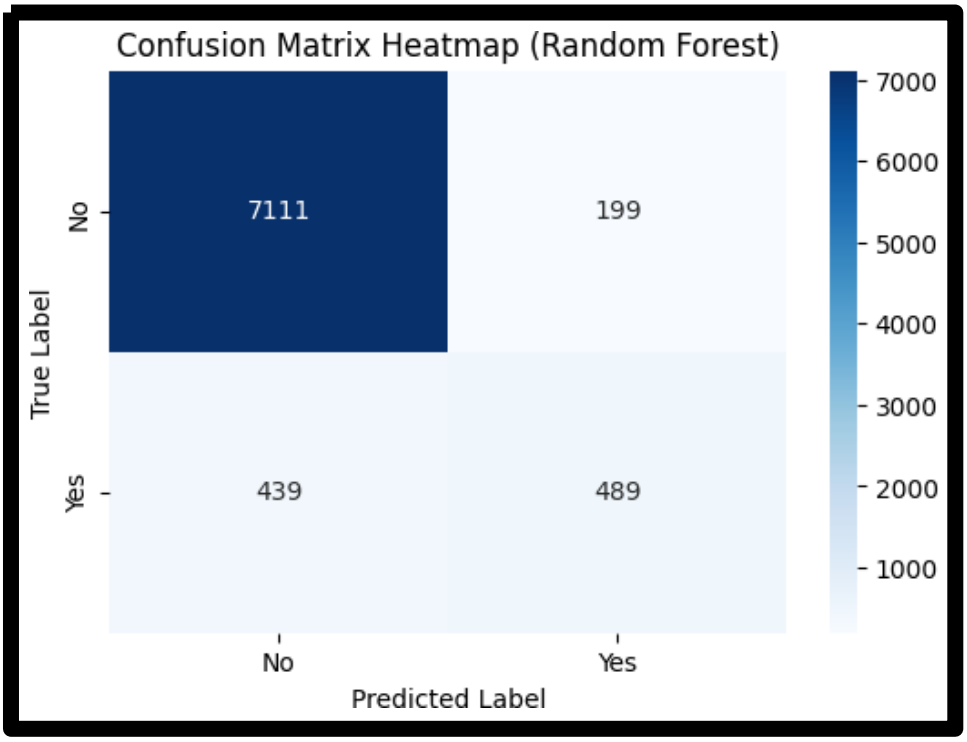
```
Confusion Matrix:

[[7111 199]

 [ 439  489]]
```

INPUT: Confusion Matrix Heatmap

```
print("\nStep 11(a): Confusion Matrix Heatmap")

import matplotlib.pyplot as plt

import seaborn as sns

plt.figure(figsize=(6, 4)) sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['No', 'Yes'],

yticklabels=['No', 'Yes'])

plt.xlabel("Predicted Label")

plt.ylabel("True Label")

plt.title("Confusion Matrix Heatmap (Random Forest)")
```

plt.show()

OUTPUT:



<u>ENTROPY</u>

INPUT: Initialization Of Random Forest Model

rf_model = RandomForestClassifier(criterion='entropy', n_estimators=200, max_depth=15,

min_samples_split=5, min_samples_leaf=2, random_state=42, n_jobs=-1)

print("Random Forest model initialized\n")

OUTPUT:

Random Forest model initialized

INPUT: Check The Accuracy

accuracy = accuracy_score(y_test, y_pred)

print(f"Overall Accuracy: {accuracy:.4f}")

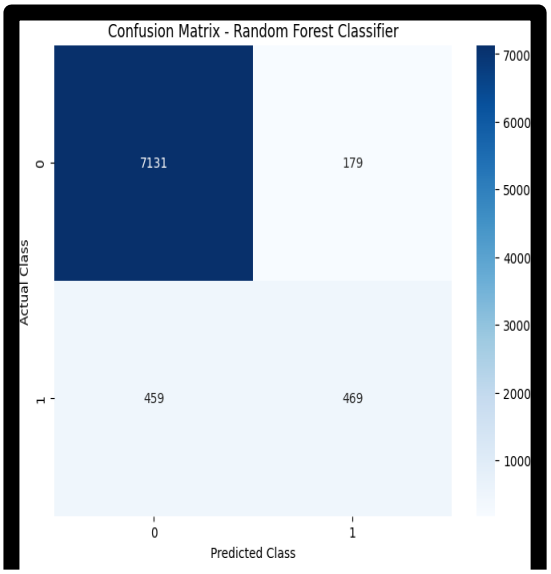
OUTPUT:

Overall Accuracy: 0.9226

INPUT: Confusion Matrix Heatmap

```
plt.figure(figsize=(8, 6))

sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=class_labels, ticklabels=class_labels)

plt.xlabel("Predicted Class")

plt.ylabel("Actual Class")

plt.title("Confusion Matrix - Random Forest Classifier")

plt.show()
```

OUTPUT:



<u>CODE FOR MULTI CLASSIFICATION</u>

INPUT: Import Libraries

```
# Core Libraries

import pandas as pd

import numpy as np

from sklearn.ensemble import RandomForestClassifier

from sklearn.model_selection import train_test_split

from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

import matplotlib.pyplot as plt
```

import seaborn as sns

print("Libraries imported successfully\n")

OUTPUT:

Libraries imported successfully

INPUT: Loading Dataset

df =pd.read_csv("seeds_dataset.txt", sep=r"\s+", header=None, names=column_names)

print("Dataset loaded successfully")

OUTPUT:

Dataset loaded successfully.

INPUT: Check the Data Shape

print("Dataset Shape:", df.shape)

OUTPUT:

Dataset Shape: (210, 8)

INPUT: Display flrst 5 rows

print(df.head())

OUTPUT:

```
First 5 Records:
    Area  Perimeter  Compactness  Length_of_kernel  Width_of_kernel  \
0  15.26      14.84       0.8710             5.763            3.312
1  14.88      14.57       0.8811             5.554            3.333
2  14.29      14.09       0.9050             5.291            3.337
3  13.84      13.94       0.8955             5.324            3.379
4  16.14      14.99       0.9034             5.658            3.562

   Asymmetry_coefficient  Length_of_kernel_groove  Class
0                  2.221                    5.220      1
1                  1.018                    4.956      1
2                  2.699                    4.825      1
3                  2.259                    4.805      1
4                  1.355                    5.175      1
```

INPUT: Data Split

Separate features and target

X = df.drop("Class", axis=1)

y=df["Class"]

```
# Train-test split

X_train, X_test, y_train, y_test = train_test_split( X, y, test_size=0.25, random_state=42, stratify=y)

print("Data Splitting Complete” )
```

OUTPUT:

Data Splitting Complete

<u>GINI</u>

INPUT: MODEL TRAINING

```
# Initialize Random Forest with GINI criterion rf = RandomForestClassifier(n_estimators=100,

criterion="gini", random_state=42, n_jobs=-1)

# Train the model

rf.fit(X_train, y_train)

y_pred = rf.predict(X_test)

print("Data Training Complete” )
```

OUTPUT:

Data Training Complete

INPUT: Model Evaluation

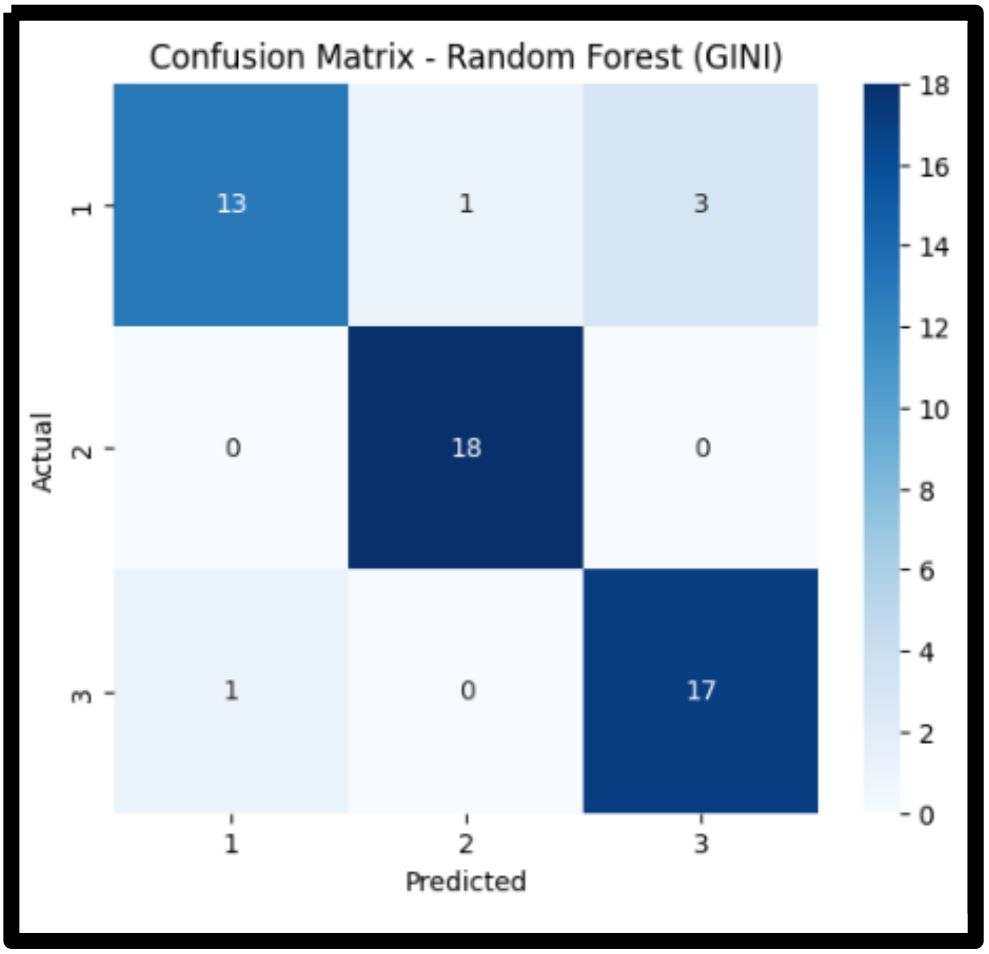
```
accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)

print("\nClassification Report:\n")

print(classification_report(y_test, y_pred))

cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 5))

sns.heatmap(cm, annot = True, fmt = “d”,cmap="Blues", xticklabels=np.unique(y),

yticklabels=np.unique(y))

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.title("Confusion Matrix - Random Forest (GINI)")

plt.show()
```

OUTPUT:

Accuracy: 0.9056603773584906

Classification Report

Class	Precision	Recall	F1-Score	Support
1	0.93	0.76	0.84	17
2	0.95	1.00	0.97	18
3	0.85	0.94	0.89	18
Accuracy			**0.91**	**53**
Macro Avg	**0.91**	**0.90**	**0.90**	**53**
Weighted Avg	**0.91**	**0.91**	**0.90**	**53**



HYPERPERAMETER(GINI):

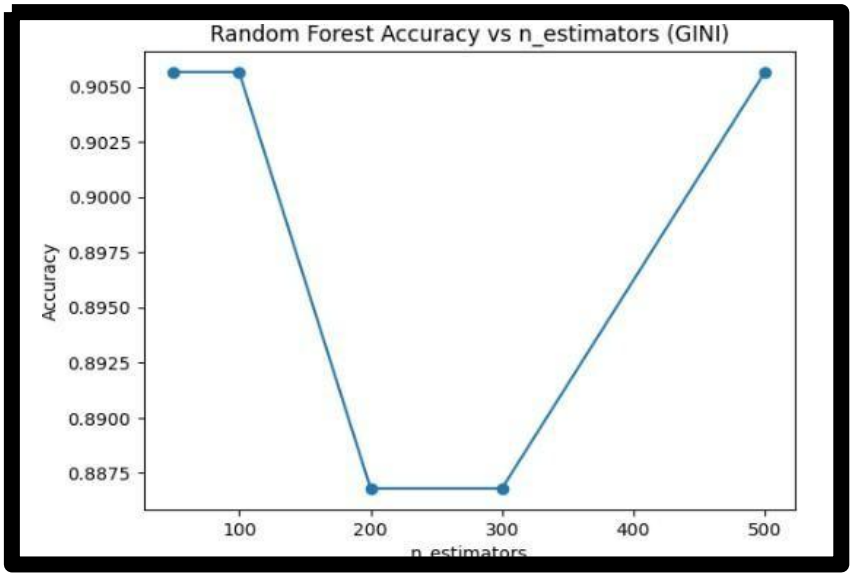
INPUT: n_estimator

```
# n_estimators values

n_values = [50, 100,200,300,500]

accuracies = []

# Train & evaluate

for n in n_values:

    model = RandomForestClassifier(n_estimators=n, criterion="gini", random_state=42)

    model.fit(X_train, y_train)

    y_pred =model.predict(X_test)

    acc = accuracy_score(y_test, y_pred)
```

```
    accuracies.append(acc)

    print(f"GINI | n_estimators={n} |  Accuracy={acc:.4f}")

# Plot graph

  plt.figure()

  plt.plot(n_values, accuracies, marker='o')

  plt.xlabel("n_estimators")

  plt.ylabel("Accuracy")

  plt.title("Random Forest Accuracy vs n_estimators (GINI)")

  plt.show()
```

OUTPUT:

```
GINI | n_estimators=50 | Accuracy=0.9057

GINI | n_estimators=100 | Accuracy=0.9057

GINI | n_estimators=200 | Accuracy=0.8868

GINI | n_estimators=300 | Accuracy=0.8868

GINI | n_estimators=500 | Accuracy=0.9057
```



ENTROPY

INPUT: Model Training

```
# Initialize Random Forest with ENTROPY criterion

rf_entropy = RandomForestClassifier(n_estimators=200, criterion="entropy", random_state=42,

n_jobs=-1)

# Train the model

rf_entropy.fit(X_train, y_train)

y_pred_entropy = rf_entropy.predict(X_test)

print("Data Training Complete” )
```

OUTPUT:

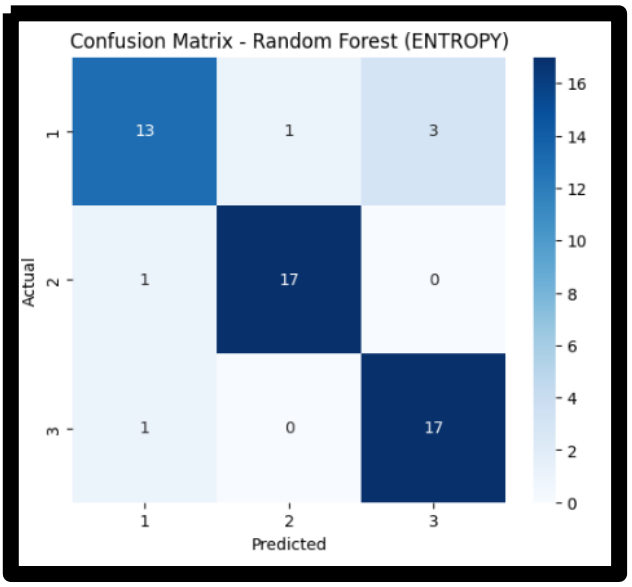
Data Training Complete

INPUT: Model Evaluation

```
accuracy_entropy = accuracy_score(y_test, y_pred_entropy)

print("Accuracy (Entropy):", accuracy_entropy)

cm_entropy = confusion_matrix(y_test, y_pred_entropy)

plt.figure(figsize=(6, 5)) sns.heatmap(cm_entropy, annot=True, fmt="d", cmap="Blues",

xticklabels=np.unique(y), yticklabels=np.unique(y))

plt.xlabel("Predicted")

plt.ylabel("Actual")

plt.title("Confusion Matrix - Random Forest (ENTROPY)")

plt.show()
```

OUTPUT:

Accuracy (Entropy): 0.8867924528301887



<u>HYPERPERAMETER(ENTROPY):</u>
OUTPUT:

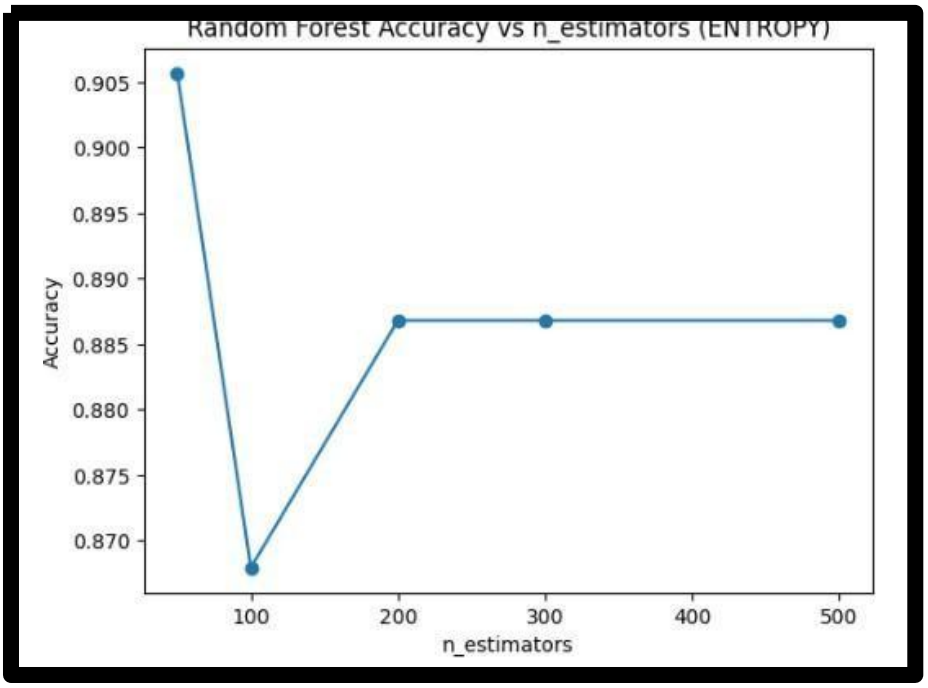
ENTROPY | n_estimators=50 | Accuracy=0.9057

ENTROPY | n_estimators=100 | Accuracy=0.8679

ENTROPY | n_estimators=200 | Accuracy=0.8868

ENTROPY | n_estimators=300 | Accuracy=0.8868

ENTROPY | n_estimators=500 | Accuracy=0.8868



Random Forest - Practical Implementation

Virtual Labs - Dayalbagh Educational Institute

Option A: Binary Classification (Bank Marketing Dataset)

Step 1: Importing Libraries

```
import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

print("Libraries imported successfully")
```

Output:

```
Libraries imported successfully
```

Step 2: Loading Dataset

 # Import necessary libraries

```
df = pd.read_csv("bank_marketing.csv", sep=';')

print("Dataset loaded successfully")
```

Output:

```
Dataset loaded successfully
```

Step 3: Data Analysis

 # Check dataset dimensions

```
print("Dataset shape:", df.shape)
```

Output:

```
Dataset shape: (41188, 21)
```

 # Preview first 5 records of the dataset

```
print("First 5 rows:")
print(df.head())
```

Output:

```
First 5 rows:
age  job        marital  education    default  housing  loan  contact    ...
0  56   housemaid  married  basic.4y     no       no       no    telephone  ...
1  57   services   married  high.school  unknown  no       no    telephone  ...
2  37   services   married  high.school  no       yes      no    telephone  ...
3  40   admin.     married  basic.6y     no       no       no    telephone  ...
4  56   services   married  high.school  no       no       yes   telephone  ...
[5 rows x 21 columns]
```

Check dataset information and types

```
print("Dataset information:")
df.info()
```

Output:

```
Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 41188 entries, 0 to 41187
Data columns (total 21 columns):
#   Column          Non-Null Count  Dtype
---  ------          --------------  -----
0   age             41188 non-null  int64
1   job             41188 non-null  object
2   marital         41188 non-null  object
3   education       41188 non-null  object
4   default         41188 non-null  object
5   housing         41188 non-null  object
6   loan            41188 non-null  object
7   contact         41188 non-null  object
8   month           41188 non-null  object
9   day_of_week     41188 non-null  object
10  duration        41188 non-null  int64
11  campaign        41188 non-null  int64
12  pdays           41188 non-null  int64
13  previous        41188 non-null  int64
14  poutcome        41188 non-null  object
15  emp.var.rate    41188 non-null  float64
16  cons.price.idx  41188 non-null  float64
17  cons.conf.idx   41188 non-null  float64
18  euribor3m       41188 non-null  float64
19  nr.employed     41188 non-null  float64
20  y               41188 non-null  object
dtypes: float64(5), int64(5), object(11)
memory usage: 6.6+ MB
```

Step 4: Data Preprocessing

Perform Label Encoding on categorical variables

```
df_encoded = df.copy()
le = LabelEncoder()

for col in df_encoded.columns:
    if df_encoded[col].dtype == 'object':
        df_encoded[col] = le.fit_transform(df_encoded[col])
        print(f"Encoded column: {col}")

print("\\nCategorical encoding completed")
```

Output:

```
Encoded column: job
Encoded column: marital
Encoded column: education
Encoded column: default
Encoded column: housing
Encoded column: loan
Encoded column: contact
Encoded column: month
Encoded column: day_of_week
Encoded column: poutcome
Encoded column: y
Categorical encoding completed
```

Separate independent and dependent variables

```
X = df_encoded.drop('y', axis=1)
y = df_encoded['y']

print("Features and target separated")
```

Output:

```
Features and target separated
```

Split data into Training and Testing sets

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f"Training set: {len(X_train)} samples")
print(f"Testing set: {len(X_test)} samples")
```

Output:

```
Training set: 32950 samples
Testing set: 8238 samples
```

Option A-i: Binary - Gini Impurity

Step 5: Model Training

Initialize Random Forest classifier with Gini criterion

```
rf_model = RandomForestClassifier(
    n_estimators=200,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    random_state=42,
    n_jobs=-1
)

print("Random Forest model initialized with Gini criterion")
```

Output:

```
Random Forest model initialized with Gini criterion
```

Fit the model on training data

```
rf_model.fit(X_train, y_train)

print("Model training completed")
print(rf_model)
```

Output:

```
Model training completed
RandomForestClassifier(max_depth=15, min_samples_leaf=2, min_samples_split=5, n_estimators=200,
n_jobs=-1, random_state=42)
```

Step 6: Model Evaluation

Generate predictions on test data

```
y_pred = rf_model.predict(X_test)

print(f"Predictions generated for {len(y_pred)} test samples")
```

Output:

```
Predictions generated for 8238 test samples
```

Calculate overall accuracy of the model

```
accuracy = accuracy_score(y_test, y_pred)

print(f"Accuracy of Random Forest model: {accuracy}")
```

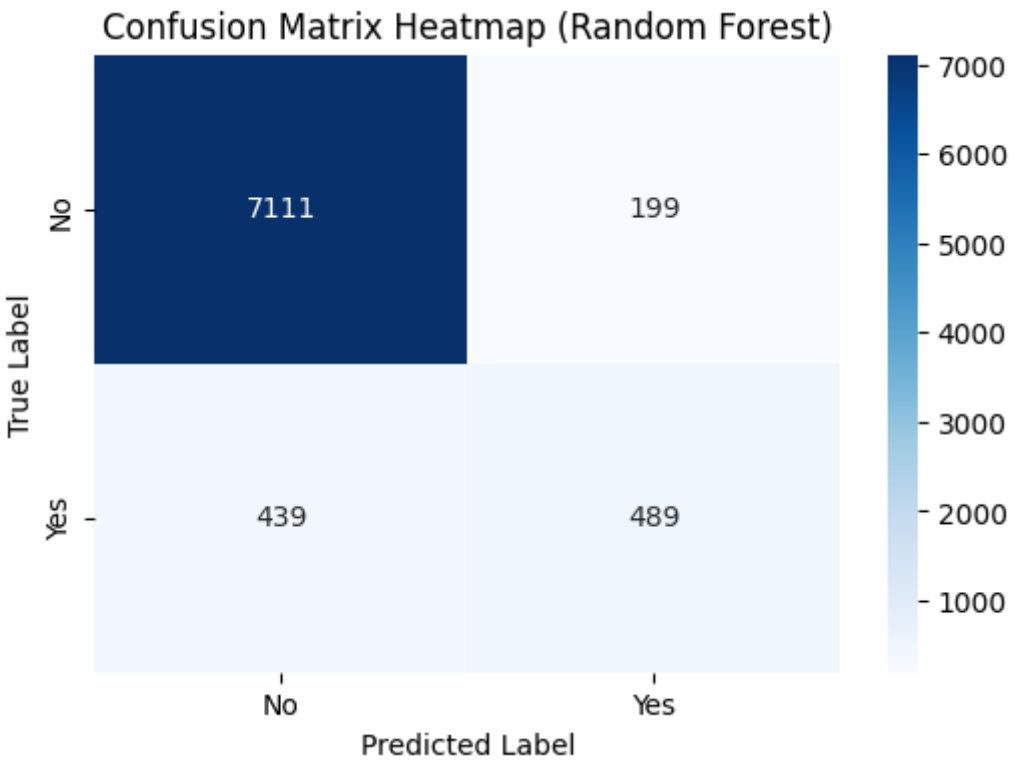
Output:
```
Accuracy of Random Forest model: 0.9225540179655256
```

Display detailed classification report

```
print(classification_report(y_test, y_pred))
```

Output:
```
Classification Report:
precision    recall  f1-score   support
0       0.94      0.97      0.96      7310
1       0.71      0.53      0.61       928
accuracy                           0.92      8238
macro avg       0.83      0.75      0.78      8238
weighted avg       0.92      0.92      0.92      8238
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 4))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['No', 'Yes'],
    yticklabels=['No', 'Yes']
)
plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix Heatmap (Random Forest)")
plt.show()
```



Step 7: Model Simulation

 # Test prediction on sample data

```
sample_data = [[56, 1, 1, 0, 0, 0, 0, 1, 0, 0, 261, 1, 999, 0, 0, 1.1, 93.994, -36.4,
    4.857, 5191.0]]

prediction = rf_model.predict(sample_data)

if prediction[0] == 0:
    print("Prediction: No (Customer will NOT subscribe)")
else:
    print("Prediction: Yes (Customer will subscribe)")
```

Output:

```
Prediction: No (Customer will NOT subscribe)
```

Option A-ii: Binary - Entropy

Step 5: Model Training

 # Initialize Random Forest classifier with Entropy criterion

```
rf_entropy = RandomForestClassifier(
    criterion='entropy',
    random_state=42,
    n_estimators=200,
    max_depth=15,
    min_samples_split=5,
    min_samples_leaf=2,
    n_jobs=-1
)

print("Random Forest model initialized with Entropy criterion")
```

Output:

```
Random Forest model initialized with Entropy criterion
```

 # Fit the model on training data

```
rf_entropy.fit(X_train, y_train)

print("Model training completed")
```

Output:

```
Model training completed
```

Step 6: Model Evaluation

Generate predictions on test data

```
y_pred = rf_entropy.predict(X_test)

print(f"Predictions generated for {len(y_pred)} test samples")
```

Output:

```
Predictions generated for 8238 test samples
```

Calculate overall model accuracy

```
accuracy = accuracy_score(y_test, y_pred)

print(f"Overall Accuracy: {accuracy:.4f}")
```

Output:

```
Overall Accuracy: 0.9226
```

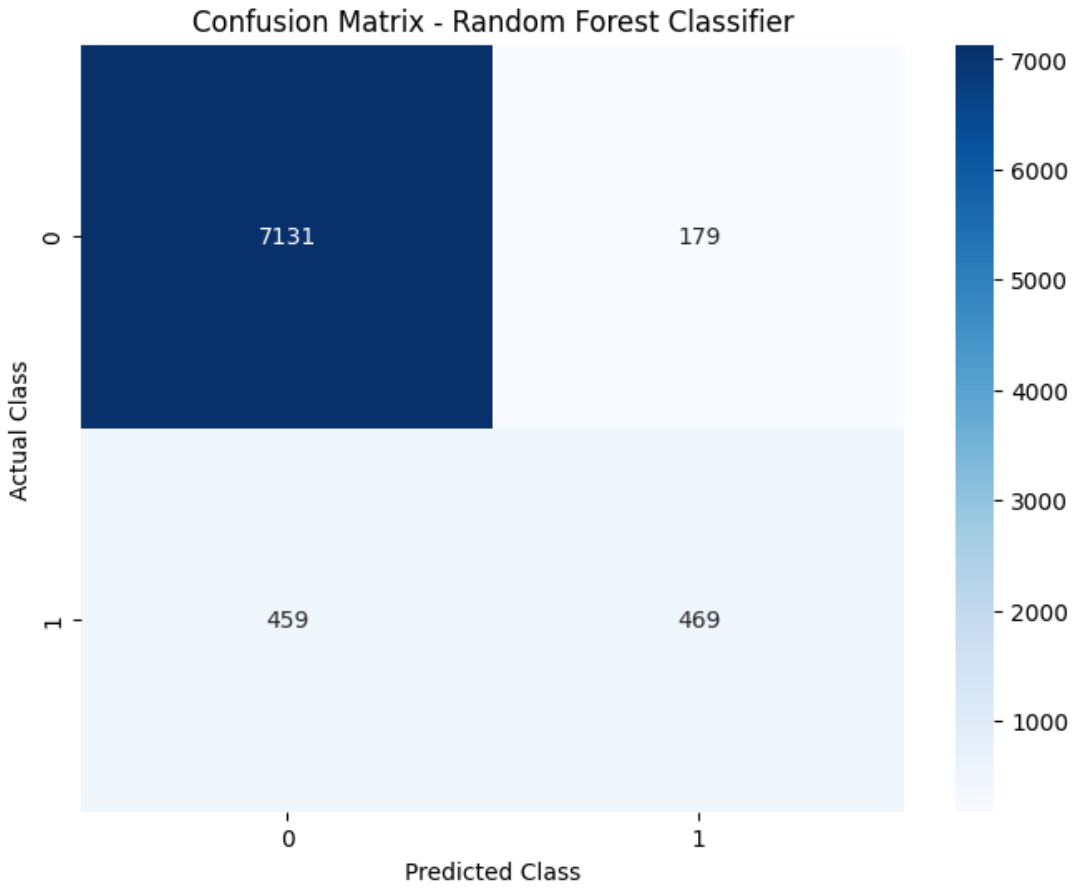
Display detailed classification report

```
print(classification_report(y_test, y_pred))
```

Output:

```
Classification Report:
precision    recall  f1-score   support
0       0.94      0.98      0.96      7310
1       0.72      0.51      0.60       928
accuracy                           0.92      8238
macro avg       0.83      0.74      0.78      8238
weighted avg       0.92      0.92      0.92      8238
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8, 6))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues',
    xticklabels=['No', 'Yes'],
    yticklabels=['No', 'Yes']
)
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("Confusion Matrix - Random Forest Classifier")
plt.show()
```



Step 7: Model Simulation

 # Test prediction on sample data

```
sample_data = [[56, 1, 1, 0, 0, 0, 0, 1, 0, 0, 261, 1, 999, 0, 0, 1.1, 93.994, -36.4,
    4.857, 5191.0]]

prediction = rf_entropy.predict(sample_data)

if prediction[0] == 0:
    print("Prediction: No (Customer will NOT subscribe)")
else:
    print("Prediction: Yes (Customer will subscribe)")
```

Output:

```
Prediction: No (Customer will NOT subscribe)
```

Option B: Multiclass Classification (Seeds Dataset)

Step 1: Importing Libraries

Import core data handling libraries

```
import pandas as pd
import numpy as np

print("Pandas and Numpy imported")
```

Output:
```
Pandas and Numpy imported
```

Import machine learning models and utilities

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

print("Scikit-learn modules imported")
```

Output:
```
Scikit-learn modules imported
```

Import visualization tools

```
import matplotlib.pyplot as plt
import seaborn as sns

print("Visualization libraries imported")
```

Output:
```
Visualization libraries imported
```

Step 2: Loading Dataset

Define dataset column names

```
column_names = [
    "Area", "Perimeter", "Compactness",
    "Length_of_kernel", "Width_of_kernel",
    "Asymmetry_coefficient", "Length_of_kernel_groove", "Class"
]

print("Column names defined:", column_names)
```

Output:
```
Column names defined: ["Area", "Perimeter", "Compactness", "Length_of_kernel",
"Width_of_kernel", "Asymmetry_coefficient", "Length_of_kernel_groove", "Class"]
```

Load Seeds dataset from local file

```
df = pd.read_csv("seeds_dataset.txt", sep=r"\\s+", header=None, names=column_names)

print("Dataset loaded from seeds_dataset.txt")
```

Output:

```
Dataset loaded from seeds_dataset.txt
```

Step 3: Data Analysis

Check dataset dimensions

```
print("Dataset shape:", df.shape)
```

Output:

```
Dataset shape: (210, 8)
```

Preview first 5 records of the dataset

```
print("First 5 rows:")
print(df.head())
```

Output:

```
First 5 rows:
Area   Perimeter  Compactness  Length_of_kernel  Width_of_kernel  Asymmetry_coeffici
Length_of_kernel_g  Class
0  15.26  14.84      0.8710       5.763             3.312            2.221               5.220
1
1  14.88  14.57      0.8811       5.554             3.333            1.018               4.956
1
2  14.29  14.09      0.9050       5.291             3.337            2.699               4.825
1
3  13.84  13.94      0.8955       5.324             3.379            2.259               4.805
1
4  16.14  14.99      0.9034       5.658             3.562            1.355               5.175
1
```

Check the distribution of target classes

```
print("Class distribution:")
print(df["Class"].value_counts())
```

Output:

```
Class distribution:
1    70
2    70
3    70
Name: Class, dtype: int64
```

Generate descriptive statistical summary

```
print("Statistical summary:")
print(df.describe())
```

Output:

```
Statistical summary:
Area        Perimeter   Compactness  Length_of_kernel  Width_of_kernel  Asymmetry_coeffici
Length_of_kernel_g  Class
count  210.000000  210.000000  210.000000   210.000000        210.000000       210.000000
210.000000          210.000000
mean   14.847524   14.559286   0.870999     5.628533          3.258605         3.700201
5.408071            2.000000
std    2.909699    1.305959    0.023629     0.443063          0.377714         1.503557
0.491480            0.818448
min    10.590000   12.410000   0.808100     4.899000          2.630000         0.765100
4.519000            1.000000
25%    12.270000   13.450000   0.856900     5.262250          2.944000         2.561500
5.045000            1.000000
50%    14.355000   14.320000   0.873450     5.523500          3.237000         3.599000
5.223000            2.000000
75%    17.305000   15.715000   0.887775     5.979750          3.561750         4.768750
5.877000            3.000000
max    21.180000   17.250000   0.918300     6.675000          4.033000         8.456000
6.550000            3.000000
```

Step 4: Data Preprocessing

 # Separate features and target variable

```
X = df.drop("Class", axis=1)
y = df["Class"]

print("Features and target separated")
```

Output:

```
Features and target separated
```

 # Split data into training (75%) and testing (25%) sets

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)

print(f"Training set: {len(X_train)} samples")
print(f"Testing set: {len(X_test)} samples")
```

Output:

```
Training set: 157 samples
Testing set: 53 samples
```

Option B-i: Multiclass - Gini Impurity

Step 5: Model Training

 # Initialize Random Forest with GINI criterion

```
rf_gini = RandomForestClassifier(n_estimators=100, criterion="gini", random_state=42)

print("Random Forest model (Gini) initialized")
```

Output:
```
Random Forest model (Gini) initialized
```

 # Train Random Forest model using GINI

```
rf_gini.fit(X_train, y_train)

print("Model training completed")
print(rf_gini)
```

Output:
```
Model training completed
RandomForestClassifier(random_state=42)
```

Step 6: Model Evaluation

 # Calculate and display GINI model accuracy

```
accuracy = accuracy_score(y_test, rf_gini.predict(X_test))

print(f"Model accuracy: {accuracy}")
```

Output:
```
Model accuracy: 0.9056603773584906
```

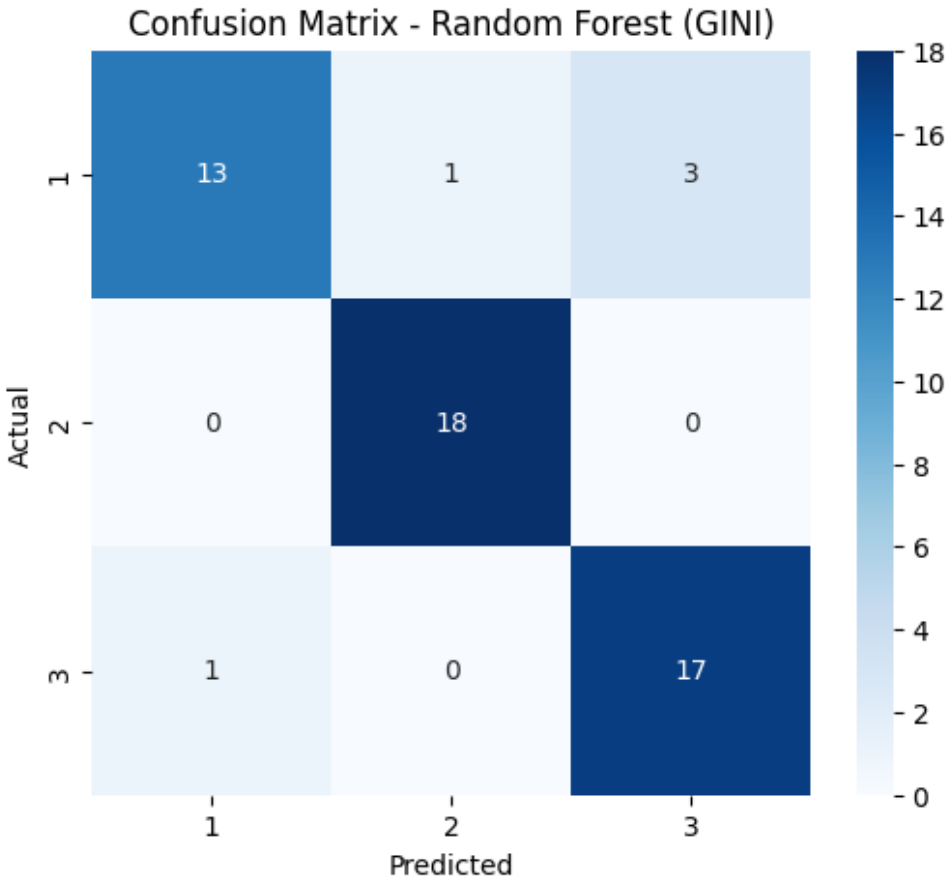
 # Display detailed classification report for GINI model

```
print(classification_report(y_test, rf_gini.predict(X_test)))
```

Output:
```
precision    recall  f1-score   support
1       0.93      0.76      0.84        17
2       0.95      1.00      0.97        18
3       0.85      0.94      0.89        18
accuracy                           0.91        53
macro avg       0.91      0.90      0.90        53
weighted avg       0.91      0.91      0.90        53
```

 # Plot confusion matrix for GINI model predictions

```
cm_gini = confusion_matrix(y_test, rf_gini.predict(X_test))
sns.heatmap(cm_gini, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix - GINI")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Step 7: Model Simulation

 # Select sample test indices representing different classes

```
sample_idx = [0, 20, 40]

print("Sample indices selected:", sample_idx)
```

Output:

```
Sample indices selected: [0, 20, 40]
```

 # Perform manual prediction and compare with true labels

```
samples = X_test.iloc[sample_idx]
true_labels = y_test.iloc[sample_idx]
preds = rf_gini.predict(samples)

for i in range(len(sample_idx)):
    print(f"Sample {i+1} Prediction: {preds[i]} (True: {true_labels.iloc[i]})")
```

Output:

```
Sample 1 Prediction: 1 (True: 1)
Sample 2 Prediction: 2 (True: 2)
Sample 3 Prediction: 3 (True: 3)
```

Option B-ii: Multiclass - Entropy

Step 5: Model Training

 # Initialize Random Forest with ENTROPY criterion

```
rf_entropy = RandomForestClassifier(n_estimators=100, criterion="entropy",
    random_state=42)

print("Random Forest model (Entropy) initialized")
```

Output:

```
Random Forest model (Entropy) initialized
```

 # Train Random Forest model using ENTROPY

```
rf_entropy.fit(X_train, y_train)

print("Model training completed")
print(rf_entropy)
```

Output:

```
Model training completed
RandomForestClassifier(criterion='entropy', random_state=42)
```

Step 6: Model Evaluation

 # Calculate and display ENTROPY model accuracy

```
accuracy = accuracy_score(y_test, rf_entropy.predict(X_test))

print(f"Model accuracy: {accuracy}")
```

Output:

```
Model accuracy: 0.8867924528301887
```

Display detailed classification report for ENTROPY model

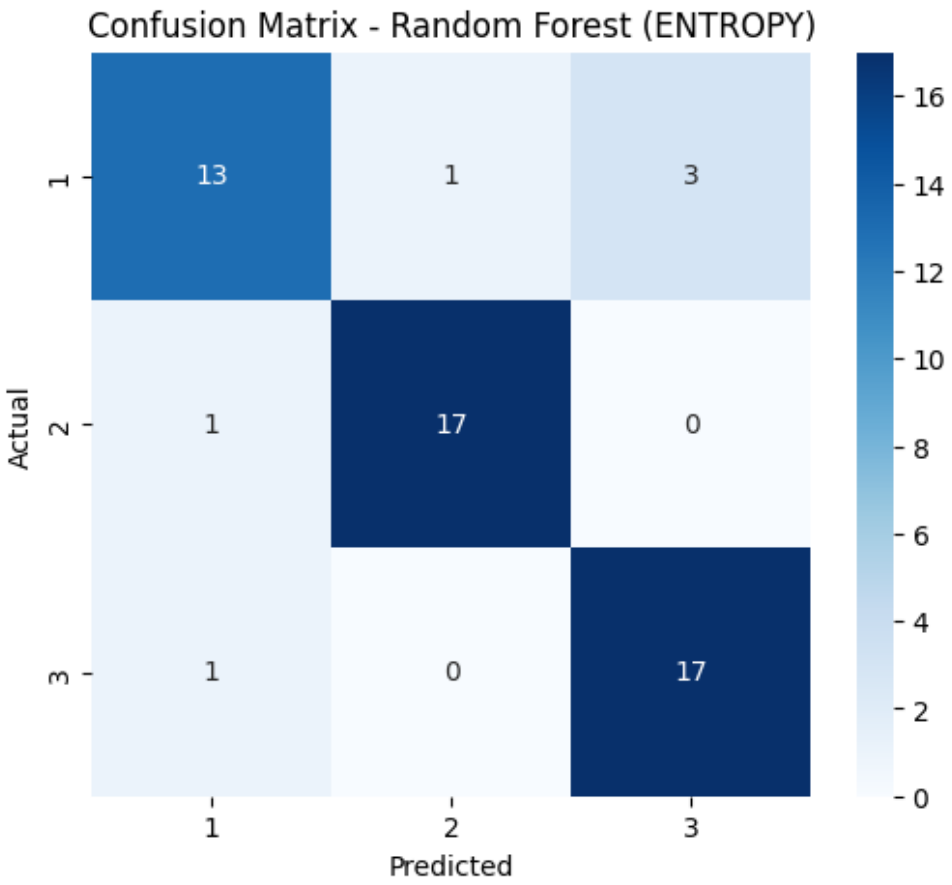
```
print(classification_report(y_test, rf_entropy.predict(X_test)))
```

Output:

```
Classification Report:
precision    recall  f1-score   support
1       0.88      0.82      0.85        17
2       0.95      1.00      0.97        18
3       0.84      0.83      0.83        18
accuracy                           0.89        53
macro avg       0.89      0.89      0.88        53
weighted avg       0.89      0.89      0.89        53
```

Plot confusion matrix for ENTROPY model predictions

```
cm_entropy = confusion_matrix(y_test, rf_entropy.predict(X_test))
sns.heatmap(cm_entropy, annot=True, fmt="d", cmap="Blues")
plt.title("Confusion Matrix - ENTROPY")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```



Step 7: Model Simulation

 # Select sample test indices representing different classes

```
sample_idx = [0, 20, 40]

print("Sample indices selected:", sample_idx)
```

Output:

```
Sample indices selected: [0, 20, 40]
```

 # Perform manual prediction and compare with true labels

```
samples = X_test.iloc[sample_idx]
true_labels = y_test.iloc[sample_idx]
preds = rf_entropy.predict(samples)

for i in range(len(sample_idx)):
    print(f"Sample {i+1} Prediction: {preds[i]} (True: {true_labels.iloc[i]})")
```

Output:

```
Sample 1 Prediction: 1 (True: 1)
Sample 2 Prediction: 2 (True: 2)
Sample 3 Prediction: 3 (True: 3)
```